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A social media analytics application of impression management and social presence theories to Twitter interaction analysis

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ABSTRACT

This study uses social media analytics to analyze Twitter users' views, opinions, and sentiments towards the Winter Olympics 2022. Impression Management Theory (IMT) and Social Presence Theory (SPT) have been utilized to analyze the imposition of the decision to practice self-censorship to sports fans, spectators, and athletes for sharing Winter Olympics content over social media channels other than Weibo. A total of 3270 tweets were collected using the combination of a keyword and a hashtag "Weibo" and "#Beijing2022". The findings suggest that Twitter users chose to remain vocal against the ban and disregarded the idea of self-censorship, as advised by Weibo. The results of sentiment analysis show overall positive polarity. After the warning from Weibo, the polarity peaked towards the negative value, and emotions such as anger, anticipation, fear, and surprise were observed. Despite a warning from Weibo, sports fans shared Winter Olympics content on Twitter, and a few links were blocked. This paper contributes to using social media in sports as a freer space to let users decide about sharing content over their preferred social media platforms.

1. Introduction

Social media has empowered people in breaking down geographical barriers by promoting e-participation, virtual presence, and the formation of online communities [1]. Pegoraro [2] considers Twitter a disruptive innovation in sport communication. Instant communication and independence to form communities have opened up new opportunities for sharing views, information and doing business. The traditional televised form of sport communication based on a one-to-many and single-medium framework has drastically changed to the participatory form of sports consumption. The fans today have greater control of sport communication through the ability of electronic-word-of-mouth experiences. Ephemeral social media empowers fans to control the duration of the messages to be available thereby, fans possess the ability to limit the reach and exposure of messages which is essential in managing relations with the brands [3–5]. Social media empowers sports fans to directly participate in building team decisions and sport-related business decisions [6,7]. Sports fans use social media to build strong relationships with athletes and sports teams, as the study by Navigate Research suggests that 67% of sports fans are more likely to use Twitter as compared to non-sport fans. The use of hashtags and seamless Twitter conversations help in enhancing online interactions enabling strangers to communicate across geological barriers [8].

The political discussions over online networks spread quickly like wildfire as they incorporate and attract larger audiences performing in-depth discussions [9]. Recent news drew the attention of Weibo, China's equivalent of Twitter which holds more than 570 million monthly active users, posted a notice on its official account and sent a direct message to all its users to implement self-censorship over posting any content relating to Beijing Winter Olympics that belonged to the broadcast rights holders. The users were strictly warned of being blocked over such activity [10]. This censorship was noticed when the People's Republic of China bagged the opportunity to host the 2022 Winter Olympics in 2015 but witnessed a sudden uproar from young biologists over the selection of an alpine ski site that is supposed to be located in the protected national nature reserve and contains many rare species of plants that include Beijing's only Shanxi orchids (*Cypripedium shanxiense*) [11]. Findings suggest that natural elements attract people more than artificial setups [12]. The posts by scientists became inaccessible just after a few hours on Weibo. This discourse was viewed as a threat to national security and hence Chinese authorities took a proactive approach to censor all the related content and media on social networks [13]. However, The Chinese Olympic Committee and the Beijing government chose to remain silent over the same [11].

Twitter users are the individuals whose perception of contact with each other in the process of communication with social media is

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analyzed in the context of shortening the social distance between them. In this paper, we have used impression management theory and social presence theory to analyze the decision of sports fans to use Twitter during the Winter Olympics and share content despite the self-censorship proposed by Weibo. Social media is a freer space that allows people to share content in an instant across the globe. Analyzing the decision of prohibiting sports fans and all related entities from sharing the content of the global mega sports event over social media platforms that are outside the control of a government is important as sports fans are the drivers of any successful sports event. The following research questions were devised to analyze the impact of the decision to impose self-censorship during the Winter Olympics:

RQ1: What is the level of enthusiasm among Twitter users for the Winter Olympics?

This question aims to estimate the enthusiasm among Twitter users through temporal statistical analysis of tweets. The comparison has been performed on the base generated over an hour each day.

H1a: Enthusiasm for the Winter Olympics and the intensity of Twitter conversations were low during the time of COVID-19.

With H1a, we hypothesize that the average discussion over all six days estimated on an hourly basis was the same. The discussion here represents the tweeting, retweeting, quoting, or mentioning behavior of Twitter users.

H1b: Weibo's rhetoric for impression building had a significant impact on Twitter conversations over sharing live content of the Olympics.

H1c: Twitter users in specific countries were tweeting heavily.

Findings by Lee [14] suggest that people who tweet between 2:00 am–3:00 am receive the highest engagements. The engagements are measured in terms of clicks, retweets, replies, and likes. H1c targets to analyze the tweeting behavior on an hourly basis in performing conversations for Winter Olympics.

RQ2: Were people all around the world disturbed over this imposed warning?

RQ2 strives to address the social presence and active participation in Twitter discussions taking place all over the world and the nations involved in it. Through this question, we aim to uncover the countries where Olympics sports fans are most vocal about Winter Olympics and Weibo. We also aim at highlighting the nations that are not at all communicating over Twitter.

RQ3: What were the sentiments towards Winter Olympics over social media?

Views and opinions disseminated in the form of tweets contain vital information [15] that can be used to estimate the level of inclination or opposition of people in the form of polarity and emotions [16,17]. RQ3 focuses on identifying the overall polarity (positive or negative) of the Twitter discussion over all six days and also the temporal variation in polarity and emotions, before and after the advice of self-censorship by Weibo.

RQ4: What were the themes of discussion on Twitter?

Thematic analysis is a qualitative research method for analyzing, identifying, organizing, describing, and reporting themes found in the dataset to answer the research questions [18]. RQ4 is purposed to find the words that have been used the most by the sports fans and made a contribution to present their sentiments.

This manuscript contains a total of 6 sections. The second section covers the extant literature in favor of social media, sports, sustainability, censorship, impression management theory, and social presence theory. The third section explains the research materials and techniques used for dataset curation and analysis. The results and analysis are presented in the fourth section. The fifth section provides a discussion of the findings of this study along with its implications for practice and policy. The conclusions drawn from this study are presented in the sixth section.

2. Literature review

This section highlights the existing literature in the domain of social media and censorship. We also highlight the recent studies that have been performed in this domain.

2.1. Twitter – A vital social media platform

Social media can be an effective tool for value co-creation and citizen participation. The governments play a pivotal role in turning the use of social media success by addressing the key issues of security, privacy, and ease of use [19]. Political content shared over social media is affected by planned and unplanned behavior. The unanimous opinion put forward by social media users is an outcome of planned behavior and not unplanned behavior [1,20]. Analysis of human behavior and values is key to sustainability planning and policy building in the era of information and communication technology (ICT), the Internet of Things (IoT), and big data that contains micro-information of every individual such as text and geolocation data [21]. Various factors determine the topic and platform an individual may choose to disseminate the information [22]. Social media users present nature to pass the information that has the highest chances of being viewed by friends or other users and create posts that promote healthy lifestyles thus providing relevant data with the viewpoint of policy and decision-making for sustainability [12,23]. Twitter is the most popular social media platform with ease to access data through hashtags or keywords with the help of Twitter APIs [24]. Twitter with Twitter API v2 aims to promote academic researchers to perform studies with access to full-archive search tweets, full-archive tweet counts, and Ads API that were earlier not available to the researchers. Any researcher accepted under the program of academic research can retrieve up to 10 million tweets per month, a massive jump of twenty times when only 500 thousand tweets could be downloaded by a normal user [25]. This information is diffused into social media channels through tweets that are composed of meta-information in the form of text, geolocation data, images, videos, and links [26]. Twitter has been used by researchers across various domains of academics such as emergency management, political science, urban planning, industrial sector, financial sector, tourism, media, retailing, manufacturing, education, and healthcare [24]. The researcher Caldevilla-Domínguez et al. [27] has stated that after COVID-19 Twitter has emerged as a citizen education tool for sustainable cities.

2.2. Censorship and governments

"Media censorship is a hallmark of authoritarian regimes" [28]. Many governments blatantly use power to censor the media of all sorts as per needs. Whether it is the mainstream media or modern Internet-supported social media [29,30]. The *Great Firewall* is one of the most powerful filtering mechanisms in the world to censor Internet content [31]. Practices of filtering sensitive terms and stringent measures of banning access to globally acknowledged social media platforms such as Facebook, Twitter, and Youtube have turned the Chinese Internet into almost an Intranet. By purposefully regulating and enforcing media censorship the governments are usually found with an intent to (1) fabricate social media posts for strategic distraction [32,33], (2) restrict unfavorable reports that directly threaten the legitimacy of the regime and local leaders [34–36] and (3) exercise the agenda-setting power of social media platforms [37,38].

Weibo, one of the most popular social media platforms in China is said to have applied thousands of humans and sophisticated software at work to track and block content that is not in harmony with the perspective of the government [39]. The Head Editor at Sina Weibo has called this censorship a 'real headache' as it requires effort to continuously monitor and change the list of dynamic sensitive topics and keywords that have to be blocked or deleted [40]. Where Weibo is

Table 1
Recent studies in the field of social media and censorship.

Article	Topic
Wong et al. [47]	Manipulate public sentiment through censorship.
Li [48]	Weibo's soft approach to suppress online activism
Vasist et al. [49]	Internet Censorship and social media monitoring important tools to curb disinformation
Wang and Dong [50]	Suppress balloon dispute
Armitage [51]	Censor on anti-wax campaign
Choli and Kuss [52]	Findings highlight the censorship of information during COVID-19 pandemic
Wong and Liang [53]	Effect of Censorship on people's perceptions of government
Luqiu and Yang [54]	Digital diplomacy in closed information systems that is, social media under heavy censorship
Li et al. [55]	Impact of internet censorship on mobile government microblogging services
Pan [41]	Strategies followed by global social media platforms to tackle Internet Censorship

commonly called "Chinese Twitter" [34,41,42]; WeChat is another popular app that is called "Chinese WhatsApp" has also gathered millions of users over the years and has been active in identifying and banning the content not in conformity with the government's rules [28,43]. The topics of corruption rumors, speculation, and political commentary have been electronically blocked with the terms "Falun Gong", Chinese characters for "princeling" and "June 4" as notable mentions. The users of WeChat who were found sharing prohibited materials were imposed with a three-step process of sanctions. If a person committed a third violation, then the associated account was closed. Tencent (the parent company of WeChat) keeps a watch on WeChat conversations that contained politically sensitive terms [43,44].

Self-censorship has become a common practice by the users of Weibo that is exercised in unique ways [43,45]. De-verification is one of the most commonly followed practices by the users of Weibo; specifically, journalists and media persons who are debarred from reporting further usually termed as "targeted killing". Another method used to sneak through the restrictions is using "*transformed*" words. Media users outsmarted the system by transforming the "sensitive words" into analogies, spoof jokes, metaphors, satires, synonyms, homophones, and homographs (Fu, 2013; [45]). As WeChat is predominantly a personal messaging app; the conversations related to political discourse are usually scant an indication that people prefer to examine self-censorship in personal conversations in the political domain [43,46].

Contrary to Weibo, Twitter is a much freer space that provides users the freedom to read and write as they prefer; making Twitter users have an appearance as more elite than an average internet user. This interpretation is justified in itself as Twitter provides a free environment to receive information and share personal views globally without censorship [40]. Jiang [38] has shown that governments perform agenda-setting through trending topics and censorship that changes the public discourse. The athletes and sports fans are integral to the popularity and success of any sporting event. The emotional attachment that sports fans present towards athletes can be estimated through the analysis of social media conversations [16]. But the rigorous measure of outright censorship imposed both on the sports fans and athletes over sharing content outside the domain of Chinese jurisdiction and warning by Weibo to its users over publishing content related to the Winter Olympics over other social media platforms [10] encouraged us to examine the views of Twitter users towards Weibo. Table 1 highlights the recent studies in the domain of censorship imposed over sharing content on social media.

2.3. Empirical research in Information Management (IM) and Information Systems (IS)

For performing quality research concise, interesting, socially relevant, and answerable research questions are necessary to be formulated. The focus of the researcher should be on formulating and answering the "Big Question" that connects to the maximum number of concerned people in the community [56]. To generate the possible questions, Impression Management Theory (IMT) [57] and Social Presence Theory (SPT) [58] two widely popular theories in the domain of IM/IS were applied to analyze the Twitter discussions. The IMT focuses on the endeavor of an actor to control or alter a depiction as perceived by the target audience [57]. The core objective of IMT is to analyze the goal-oriented nature of individuals and organizations that attempt to change or control peoples' perceptions [59]. In this study, IMT targets the impression management tactics followed in social media. Impression management tactics apply to social media at both individual and organizational levels [60]. Weibo in this study is the organization that has tried to manage and control the opinions of people through the promotion of self-censorship. SPT on the other hand focuses more on examining and explaining the authenticity of the communication. The social presence of people is perceived through the media and communication tasks, without considering the social factors of the communication process [58]. Social presence through online interaction and collaboration helps in reducing the psychological distance between the participants and improves the efficiency of interaction and communication [61,62].

3. Research materials and techniques

For performing this research, the availability of quality data is important that is based on the accuracy of decisions about data collection and measurement [63]. Tweets from Twitter were used as a potent source of information that was collected and processed by following processes.

3.1. Fetching the tweets

For this study, we took to Twitter for data collection as it is one of the most globally renowned social media platforms where information is spread in the form of tweets. Twitter provides users with an easy-to-use application programming interface (API), Twitter API "tweepy" that can be used to collect and download publicly available tweets. The tweets are composed of meta-information based on parameters such as tweet text, date, time, location, screen name, hashtags, user name, tweet id, and more [17]. The combination of keywords and hashtags provides the tweets that contain those words. The keywords can be simple words used in any language. The hashtags are the preferred choice of Twitter users, prefixed with the "#" symbol hashtags contain a great deal of information that is relevant for categorizing content [3,17,64]. This encouraged the authors to use hashtags for the purpose and fetch Weibo-specific tweets. The combination of keywords "*weibo*" and hashtag "*#Beijing2022*" were used in conjunction with AND logic [65]. The AND logic restricts that both values are required to be present in the tweet text. This makes the tweepy fetch the tweets that contain both "*weibo*" and "*#Beijing2022*". This technique saved a lot of time and effort that would have been wasted in removing unwanted tweets. As language is a barrier to the understanding of a text so, the search for tweets was confined to the English language only. After satisfying all the latter parameters tweepy retrieved a total of 3270 tweets that span over a period of six days from Jan 26, 2022 to Jan 31, 2022.

3.2. Preprocessing the data

Preprocessing is the most important phase of data analysis. In this study, the data that needs to be handled is in the form of text. Text data

that represents the thoughts and views of the writers usually contains noise may be in the form of misspelled words or slang that may alter the meaning of the sentence. Punctuations, stop words, web links, special characters, and whitespaces (extra blank spaces between the words or sentences) all need to be eliminated to keep the focus on the meaningful words in the sentence [17]. As this study is being conducted in India, the tweets have been collected based on Indian Standard Time (IST) and were ordered based on date and time [66]. The step of preprocessing the text and date in the data ensures the maintenance of high-quality and meaningful features that are present in the form of text, date, and time.

3.3. Research techniques

For any research to be valid and consistent with its results, requires the application of robust and appropriate techniques [67]. In this study, we have applied quantitative and qualitative research techniques to find the answers to the aforementioned research questions and research hypothesis [56,68]. To perform the temporal analysis of the tweets, ANOVA and Tukey's HSD have been applied [69]. The significant P-value of two-way ANOVA used to perform temporal statistical analysis of tweets indicates that there exists at least one pair, between which significant statistical difference exists; whereas, the F-value in ANOVA shows the variability between the groups of observations within the groups [69,70]. To find the pair that results in a significant difference, posthoc test (multiple comparisons) is applied. Tukey's HSD is a post-hoc testing method that is applied to find out the difference between the mean of all the possible pairs over a studentized range of distribution using Statsmodels, a python package specifically built for statistical analysis of the data [69,71]. The data visualization libraries Seaborn [72,73] and Matplotlib [74] available in Python programming language were used for plotting barplot, swarmplot, and lineplot with 95% confidence interval to perform the analysis [75].

The dynamic nature of the Internet in terms of volume, velocity, and variety of opinion-rich available through social media channels has made research in the domain of sentiment analysis a vital source of generating meaningful insights to help decision support and domain analysis with targeted information [17,76]. Text mining is an important task involved in sentiment analysis which is a text classification task that enables a data scientist to understand the views, opinions, and emotions of individuals disseminated in the form of text over social media [16,64,76]. This study follows a tidy approach of text mining to perform the sentiment analysis which includes polarity analysis and emotion analysis [77]. Text mining for sentiment analysis and identifying themes of Twitter discussions has been accomplished using the "Tidyverse" which is a collection of R-packages (for R programming language) that have been specifically designed for data science [78]. The sentiment analysis of each tweet was performed and a graphical representation of change in sentiments with time was done using "ggridges" library which is an extension to ggplot2 an R-package for data visualization [78,79]. Fig. 1 lists the complete research procedure followed to conduct this research. The research has been performed in three phases namely Fetch Phase, Pre-processing Phase, and Analysis Phase.

4. Results and analysis

This section is divided into two parts. In the first section, we used statistical and sentiment analysis techniques to critically examine the Twitter dataset and discover the answers to the research questions. In the second section, we have applied some of the commonly used social media analytics techniques to unearth hidden information from the users' content or tweets.

4.1. Temporal analysis

To find the answers to RQ1, we performed the temporal analysis of the tweets over six days. In RQ1, we try to analyze the use of social media by Olympic fans and spectators for the Winter Olympics. To validate our claims, we statistically validate H1a, H1b, and H1c.

Table 2
Results of Tukey HSD.

Multiple comparison of means - Tukey HSD, FWER = 0.01						
group1	group2	meandiff	p-adj	Lower	Upper	Reject
26	27	−50.5417	0.001	−86.9647	−14.118	True
26	28	−65.0417	0.001	−101.4647	−28.6187	True
26	29	−53.0417	0.001	−89.4647	−16.6187	True
26	30	−67.5833	0.001	−104.0063	−31.1603	True
26	21	−65.25	0.001	−101.673	−28.827	True
27	28	−14.5	0.7195	−50.923	21.923	False
27	29	−2.5	0.9	−38.923	33.923	False
27	30	−17.0417	0.5825	−53.4647	19.3813	False
27	31	−14.7083	0.7082	−51.1313	21.7147	False
28	29	12.0	0.8541	−24.423	48.423	False
28	30	−2.5417	0.9	−38.9647	33.8813	False
28	31	−0.2083	0.9	−36.6313	36.2147	False
29	30	−14.5417	0.7172	−50.9647	21.8813	False
29	31	−12.2083	0.8429	−48.6313	24.2147	False
30	31	2.3333	0.9	−34.0897	38.7563	False

H1a: Enthusiasm for the Winter Olympics and the intensity of Twitter conversations were low during the time of COVID-19.

For H1a, we applied one-way ANOVA using Statsmodels based on days that compared the count of tweets for every hour. The results of ANOVA at 95% confidence came out with F-Value = 4.5665 and p = 0.00121. As per the decision rule, H1a was rejected as the level of significance denoted by the p-value was less than 0.05. Latter results indicated that there was a significant difference between one or more groups of days in regard to the discussion on Twitter.

To find the group of days that resulted in the significant difference among the hourly discussions on Twitter, we performed multiple comparisons of means as shown in Table 2 and plotted Tukey's HSD diagram as shown in Fig. 2. Fig. 2 clearly shows the comparison of discussions of all the days with one another.

Inference: Jan 26 is the date that resulted in the significant difference in an overall discussion which is rejected when compared to all other dates as listed in Table 2, as it received a P-value less than 0.05 in all the comparisons, whereas, all other pairs are statistically significant.

H1b: Weibo's rhetoric for impression building had a significant impact on Twitter conversations over sharing live content of the Olympics.

The P-values listed in Table 2 for all the days except Jan 26 clearly show that there were no statistical differences in the frequency of discussions. We performed the statistical analysis of tweets on an hourly basis for each day. Table 2 lists all the values for corresponding statistical measures. This enabled us to create Fig. 3 which is the boxplot through which we examined the hourly generation of tweets globally.

From Fig. 3, it is very much clear that only on Jan 26, there were significant variations in Twitter discussions as the box is longer in length representing a high level of dispersion of tweet counts with a maximum value that reached 280 tweets during an hour. Swarmplot plotted along with the boxplot makes it easy to examine that for most days the number of tweets on an hourly basis was quite similar. Thus, making the quartiles for most of the days in the boxplot coalesce from Jan 27 to Jan 31. From the latter discussions, it can be inferred that there was not much dispersion or variation of tweet counts over the hours during these days. This indicates that Twitter users enjoy a greater level of independence and are not swayed by the warning of a private organization (see Table 3).

H1c: Twitter users in specific countries were tweeting heavily.

Fig. 4 is composed of a combination of three plots namely barplot, swarmplot, and lineplot. The tick values on the y-axis represent the count of tweets and the tick values on the x-axis represent the hours as per the 24-h scale. The bars in the barplot start from the value 0 and

Fig. 1. Research procedure.

the highest point marks the mean value of the count of tweets for all the six days generated over a particular hour. The vertical black line in the middle of every bar shows the level of variation of values and uncertainty around the estimated average value. The swarmplot shows the count of tweets generated for all six days during a particular hour.

The lineplot connects the head of bars in the barplot which are the mean values of the count of tweets for all six days for a specific hour and shows the 95% confidence in the shaded region along the lineplot. The lineplot makes it noticeably clear that the overall discussion over ''Weibo and Winter Olympics'' is confined to a particular region of the world which resulted in massive variation in the generation of tweets during particular hours for all the days.

4.2. Geo-location analysis

RQ2 aims at finding the nations that were most vocal and presented resentment or views towards the promotion of self-censorship by Weibo. For this purpose, we plotted the geo-heatmap and marked the nations based on the count of occurrence in the dataset shown in different intensities of green color. The darker shades of green represent